1. Why many tables?

- Store each fact only once
- Separate shelves: books, members, loans
- Linked together by ID numbers

```
SELECT * FROM members;
```

member_id	name	city	joined	email
1	Aisha	Austin	2023-01-15	aisha@mail.com
2	Ben	Denver	2023-03-02	NULL
3	Carlos	Austin	2024-06-20	carlos@mail.com
4	Dana	Boston	2024-09-10	NULL
5	Eli	Denver	2025-02-01	eli@mail.com

2. Keys link the shelves

- PRIMARY KEY = unique ID of each row
- FOREIGN KEY = points to another table
- Like a string between two shelves

```
SELECT loan_id, book_id, member_id
FROM loans;
```

loan_id	book_id	member_id
1	1	1
2	3	1
3	2	2
4	4	3
5	1	3

3. INNER JOIN

- Connect two tables ON matching keys
- Keeps only rows that match
- Use table nicknames: loans AS l

```
SELECT m.name, l.loan_date FROM loans AS l
JOIN members AS m
ON l.member_id = m.member_id;
```

name	loan_date
Aisha	2026-01-05
Aisha	2026-02-10
Ben	2026-01-20
Carlos	2026-03-01
Carlos	2026-04-02

4. LEFT JOIN + IS NULL

- LEFT JOIN keeps everyone from the left table
- No match? Columns become NULL
- WHERE … IS NULL finds who never did something

```
SELECT m.name FROM members AS m
LEFT JOIN loans AS l
ON m.member_id = l.member_id
WHERE l.loan_id IS NULL;
```

name
Eli

5. Practice free + what's next

- sqliteonline.com, db-fiddle.com, sqlime.org (choose SQLite)
- sqlbolt.com for free lessons
- Paste the setup from the practice guide, click Run
- Next, Part 4: subqueries, CASE, UNION, NULLs